

GraphCURE: Grounded Cross-Expert Uncertainty Routing for Evidence-Aware Multimodal Fact Checking

Anonymous ACL submission

Abstract

Multimodal fact-checking systems are vulnerable to two coupled errors: retrieved evidence can be incomplete, and a verifier optimized for a verdict can remain overconfident when the evidence does not warrant one. We study this problem in a fixed-corpus MOCHEG protocol, isolating verifier and decision-policy effects from live-web variation. We introduce GRAPHCURE (Grounded Cross-Expert Uncertainty Routing), a framework that combines a direct verdict expert with a grounded rationale expert. Its asymmetric epistemic-deferral policy preserves the direct verdict by default and emits NEI only when the grounded expert detects insufficient evidence with frozen high confidence. On frozen validation, the full policy improves Macro-F1 from 0.6920 to 0.7112 (+1.92 points; paired-bootstrap $P(\Delta > 0) = 0.9975$). We report broad component, top- K , threshold, subgroup, efficiency, and negative-ablation analyses. A validation-selected eight-model heterogeneous ensemble reaches 0.5551 Macro-F1 on the raw official P1 test ($n = 2,442$), exceeding the 0.5400 reported by the directly comparable AMuFC baseline; a one-model disagreement-distilled student reaches 0.5494. We distinguish raw official and deduplicated strict tracks, never use test labels to select confirmatory policies, and identify an unreconciled B18B routing artifact as a pre-submission TODO.

1 Introduction

Automated fact checking is commonly decomposed into evidence retrieval and claim verification (Guo et al., 2022; Thorne et al., 2018). Modern systems increasingly retrieve heterogeneous evidence and use large language models, but two assumptions remain fragile: that all retrieved passages should be consumed by the verifier, and that an expert trained for verdict prediction is also calibrated about evidence sufficiency. These assumptions are particularly consequential for multimodal

fact checking, where evidence may be noisy, missing, duplicated, or distributed across modalities.

MOCHEG (Yao et al., 2023) provides a useful setting because it couples three-way truthfulness labels with retrieved textual and visual evidence and explanations. Existing MOCHEG systems model cross-modal interactions directly, for example with hypergraph transformers (Pang et al., 2025). However, comparisons are difficult when systems use different retrieval assumptions. We therefore first establish a fixed-corpus, text-retrieval protocol and then study the verifier and routing decisions under identical evidence.

Our central observation is that two verifiers have complementary error profiles. A direct verifier is strong when evidence appears decisive but can overcommit under missing evidence. A grounded expert, trained with structured explanations, is more sensitive to insufficiency but is not uniformly better. We turn this complementarity into a selective policy rather than averaging the experts. The direct expert remains the default; only a high-confidence grounded NEI signal can defer the decision.

Our contributions are:

- GRAPHCURE, a reproducible protocol-separated framework in which grounded explanations act as an evidence-sufficiency sensor rather than a symmetric second classifier;
- an asymmetric cross-expert decision rule that isolates the contribution of NEI deferral from ensembling, confidence routing, and grounded-only prediction;
- a fully auditable empirical ledger spanning B1–B19, including negative results, train-only OOF controls, frozen validation ablations, raw official test, and strict robustness evaluation; and
- B19 disagreement-aware knowledge distillation, a compute-efficient single-model alter-

native that transfers complementary teacher behavior without inference-time routing.

We do not claim that every number in the internal development history is directly comparable. Official-test claims are restricted to frozen, auditable artifacts; unresolved discrepancies are marked in Section 11.

2 Related Work

Fact verification and evidence sufficiency. FEVER established evidence-supported three-way verification at scale (Thorne et al., 2018); LIAR studied short political claims (Wang, 2017); and MultiFC broadened evaluation across sources and domains (Augenstein et al., 2019). These benchmarks make clear that label prediction and evidence sufficiency are not identical. Surveys consequently treat retrieval, evidence selection, reasoning, and calibration as coupled but distinct stages (Guo et al., 2022). UKP-Athene illustrates the role of multi-sentence textual entailment in evidence-based verification (Hanselowski et al., 2019).

Retrieval, selection, and reasoning. Evidence retrieval can dominate downstream verification (Zheng et al., 2024). RAV combines retrieval with verification, while RAFTS synthesizes contrastive arguments from retrieved evidence (Yue et al., 2024). Complex-claim verification decomposes claims before retrieving web evidence (Chen et al., 2024a). These approaches study different evidence regimes from our P1 protocol: they motivate the separation between fixed-corpus evidence selection and live-web retrieval rather than serving as directly interchangeable leaderboard rows.

Multimodal fact checking. MOCHEG introduced end-to-end multimodal fact checking and explanation generation (Yao et al., 2023). Recent systems pursue different ways to aggregate evidence: LVLM4FV uses large vision-language models with reranked evidence (Tahmasebi et al., 2024); Meta-SumPerceiver creates claim-specific multimodal summaries (Chen et al., 2024b); HGTMF models high-order interactions with hypergraphs (Pang et al., 2025); MEVER jointly addresses retrieval, verification, and explanations (Zhang et al., 2026); and M-RAV augments cross-modal evidence alignment for zero-shot verification (Luu et al., 2027). AMuFC adaptively determines whether visual evidence is needed (Jung et al., 2026), while Entailed Opinion stresses entailment-oriented reason-

ing (Kumar et al., 2025). GRAPHCURE shares the adaptive-evidence motivation but studies an observable uncertainty-routing problem between two differently trained verifiers under an explicitly controlled P1 evidence protocol.

3 Task Definition and Protocols

Given a claim c and retrieved evidence $E = \{e_j\}_{j=1}^K$, predict $y \in \{\text{SUPPORTED}, \text{REFUTED}, \text{NEI}\}$. The principal P1 protocol uses MOCHEG’s fixed corpus and upstream text retrieval; it is not an open-web evaluation. The raw official test contains 2,442 claims. A strict robustness track removes eight verbatim cross-split duplicates and contains 2,434 claims. We report both tracks separately. Development validation contains 1,456 claims.

All routing thresholds are selected on validation or train-only out-of-fold data. Test labels are never used for selecting a seed, threshold, or policy in the confirmatory result. Open-web experiments in Phase C are not mixed into P1 tables.

4 Method

4.1 Direct and grounded experts

The direct expert f_D receives claim c and top- K retrieved passages E_K and predicts a three-way distribution $p_D(y | c, E_K)$. It is trained with ordinary verdict cross-entropy. The grounded expert f_G shares the backbone and input format but is trained with teacher-distilled structured explanations. Each teacher target contains: (i) a verdict, (ii) selected key-evidence identifiers, (iii) sufficiency/polarity fields, and (iv) an explicit missing-information field. Hence the extra supervision makes insufficiency observable in the target representation without requiring the model to generate an explanation at inference.

For a training example i , let z_i be the verdict target and r_i the structured rationale target. The two expert objectives are

$$\begin{aligned}\mathcal{L}_D &= - \sum_i \log p_D(z_i | c_i, E_i), \\ \mathcal{L}_G &= \mathcal{L}_D + \lambda_r \mathcal{L}_{\text{rationale}}(r_i).\end{aligned}\tag{1}$$

The project records teacher outputs, manifests, retrieval signatures, and prompt-format hashes. Gold evidence is used only for retrieval diagnostics, never injected into P1 validation or test prompts.

4.2 Asymmetric epistemic deferral

The policy uses the direct expert as an anchor:

$$\hat{y} = \begin{cases} \text{NEI}, & \arg \max_y p_G(y) = \text{NEI} \wedge p_G(\text{NEI}) \geq \tau, \\ \arg \max_y p_D(y), & \text{otherwise.} \end{cases} \quad (2)$$

We use $\tau = 0.49$, selected on validation and frozen thereafter. This rule is intentionally asymmetric: the grounded expert is not permitted to overwrite a direct Supported or Refuted decision with another decisive class. It can only defer when it detects insufficiency. This preserves the direct verifier’s decisive-class calibration while using the grounded branch as an epistemic sensor. It is therefore distinct from posterior averaging, symmetric confidence routing, and the oracle routing upper bound.

4.3 Disagreement-aware distillation

B19 is a secondary single-model variant. Let q_D and q_G be frozen teacher distributions. For agreement cases, the distillation target is their average; for disagreement cases, the train-only held-out outcome identifies the teacher target during cross-fitted training. The student minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{KD}} T^2 \text{KL}(q \| p_S) + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}}, \quad (3)$$

where q is the selected teacher target and T is the distillation temperature. A counterfactual term compares an evidence set with a version in which teacher-identified key passages are omitted. B19 is reported as a compute-efficient secondary contribution, not as a replacement for the main B18B routing analysis.

4.4 Validation-selected heterogeneous ensemble

The confirmatory B18A test system is a complementary, validation-selected ensemble: five frozen B1 direct seeds plus the three best B18A grounded seeds ranked on validation. The composition $\{100, 87, 42\}$ is fixed before the official test is evaluated. Its prediction is the arithmetic mean of member verdict probabilities. We also report an unpruned ten-member ensemble (all five direct plus all five grounded seeds) as a no-subset-selection control. This is important because B18A ensemble selection and B18B routing answer different questions: the former is the principal P1 benchmark system; the latter is the main decision-policy ablation.

Split/protocol	Claims	Labels used for selection	Role
Train	11,631	train only	optimization
Validation	1,456	validation	frozen policy selection
Official test	2,442	no	primary benchmark
Strict test	2,434	no	duplicate-robustness

Table 1: Protocol sizes. Official and strict test results are not pooled.

5 Experimental Setup

5.1 Data and preprocessing

MOCHEG contains 15,601 English claims from PolitiFact and Snopes with textual, visual, and social evidence (Yao et al., 2023). Our P1 experiments use the project manifest, a fixed MOCHEG text corpus, Qwen3 dense retrieval, and reranking. We retain the resulting evidence order and use $K = 5$ passages unless an evidence ablation changes K . No gold evidence is injected at validation or test. The open-web C-series snapshot is intentionally excluded from P1 effectiveness tables: it tests a different deployment regime.

5.2 Models and metrics

All verifiers use Qwen3-4B-Instruct-2507 with LoRA adapters. The direct anchor uses five seeds $\{13, 21, 42, 87, 100\}$; B18A uses grounded-rationale distillation; B18B consumes the frozen direct and grounded predictions. The representative runs use LoRA rank 16, LoRA scaling 32, dropout 0.05, learning rate 10^{-4} , AdamW weight decay 0.01, warm-up ratio 0.05, three epochs, per-device batch size 1, and gradient accumulation 16. Evidence is truncated to 2,200 characters for the standard grounded prompts; exact run configurations are retained in each JSON summary.

We report accuracy, class F1, Macro-F1, expected calibration error (ECE) when available, paired percentile bootstrap intervals, and exact McNemar tests. Bootstrap samples are drawn at the claim level with 10,000 iterations. We call $P(\Delta > 0)$ a bootstrap probability of superiority, not a frequentist p -value. All test table deltas compare predictions with matched claim IDs.

5.3 Baselines and comparison discipline

Baselines include direct-only, grounded-only, equal posterior averaging, symmetric confidence routing, no-asymmetric-NEI routing, disagreement-only routing, max-confidence routing, and oracle

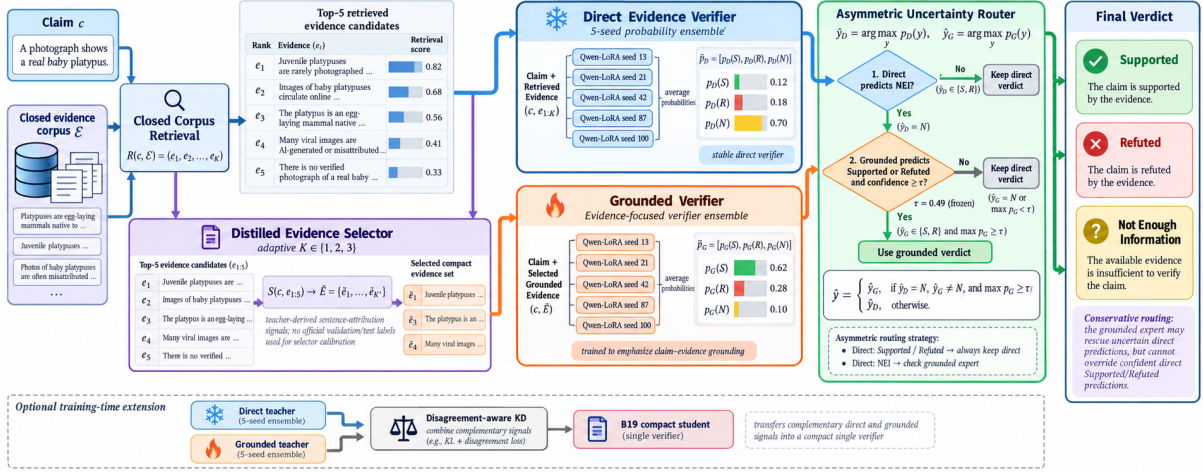


Figure 1: Overview of GRAPHCURE. The claim and retrieved evidence are processed by (i) a direct verdict expert and (ii) a grounded rationale expert. The frozen asymmetric gate may only defer a decision to NEI; otherwise it preserves the direct prediction. The lower B19 branch distills complementary expert behavior into a single student.

routing. Literature scores are reported in a separate protocol-aware table. MOCHEG, HGTMFC, AMuFC, MEVER, M-RAV, and open-web systems do not necessarily share evidence, split, modality, or metric definitions; therefore a row is not treated as a direct leaderboard comparison unless its protocol is explicitly aligned.

5.4 Promotion gates and split discipline

Every development intervention is governed by a predeclared promotion gate. From B6 onward, the project uses duplicate-family-safe train-only folds, fixed training epoch, compute-matched controls, and fresh confirmation folds. A screen-only gain therefore cannot become an official-test claim. The B18A ensemble is selected on validation and applied once to test. The post-hoc $\tau = .60$ sensitivity result is reported separately and never used as the main confirmatory result. This distinction is central to interpreting the long experimental record in Table 2.

6 Main Results

6.1 Research trajectory and negative evidence

The project did not proceed as a sequence of monotonically improving models. B2–B17 were registered hypothesis tests designed to explain or surpass the strong B1 anchor. Most were closed after fresh-fold confirmation or matched-control failure. Reporting them is important: it identifies which apparently promising mechanisms do *not* generalize and narrows the evidence-based explanation for

B18A/B18B.

6.2 Frozen validation and ablations

Broader prior-work inventory. Table 4 lists twelve relevant MOCHEG system rows. It is deliberately *not* a single leaderboard: a reported F-score is preserved as F-score rather than silently renamed Macro-F1; filtered tests, gold evidence, or alternate preprocessing are marked. Only rows labelled “aligned raw P1” support an apples-to-apples point-estimate comparison.

B18B is the strongest registered validation variant. Relative to direct-only it gains 1.916 Macro-F1 points and 1.442 accuracy points. Removing the asymmetric NEI rule reduces Macro-F1 by 0.677 points, whereas equal averaging loses 1.843 points. The grounded-only branch is useful (+0.915 over A0) but remains 1.001 points below A8. These comparisons support the claim that the gain is due to the asymmetric decision interface rather than merely adding a second model.

6.3 Official test results and leakage control

The frozen B18A heterogeneous ensemble reaches Macro-F1 0.55507 and accuracy 0.57535 on raw P1 official test, compared with the B1 five-seed baseline of 0.54531 and 0.56798. The paired bootstrap probability of improvement is 0.9839 with CI [+0.00097, +0.01850]. Compared with AMuFC’s reported raw-P1 point estimate (Macro-F1 .5400; accuracy .5460), this is a +1.507 Macro-F1-point and +2.935 accuracy-point difference. This is a

Phase	Scope	Key observed result	Decision and scientific finding
B1	P1 official	0.5680 accuracy / 0.5453 Macro-F1 (five-seed ensemble)	Frozen direct-verdict anchor; modern dense retrieval + Qwen3 verifier establishes the benchmark.
B2	visual/sufficiency	SV screen +0.0027 MF1; confirmation +0.0006 $\pm$ 0.0074	Visual relevance did not reliably become claim-image stance improvement.
B3	source robustness	qrel-absent MF1 +0.1004; overall MF1 -0.0021	Reweightings lacked source safety.
B4	sentence evidence	Recall@8=.8661; verifier MF1 -0.0270	Atomic sentences lose discourse context.
B5	context packet	5-seed gain +0.0039 $\pm$ 0.0086	Packet and article evidence are complementary but unstable.
B6	hierarchy / PCGrad	auxiliary ensemble MF1=.7080; $P(\Delta > 0) = .9354$ vs control	Auxiliary signals transfer, but not robustly enough for promotion; PCGrad did not fix it.
B7-B8	frozen routing / prior	router +.00460; logit bias +.00216	Small gains, insufficient bootstrap evidence.
B9-B11	seed calibration	B9 confirmation +.00398; cross-fit calibration +.00345	Seed variance and provenance stacking did not generalize safely.
B12-B13	constraints/curriculum	B12 +.00538 vs control but -.01376 vs anchor	Long auxiliary training induces drift; blend was not independently confirmed.
B14-B15	NEI / value gate	NEI escape +.00209; value gate +.00380	Evidence availability matters, but observable utility features were too weak.
B16-B17	counterfactual verdict	screen +.01654; confirmation -.00104 vs matched control	Counterfactual supervision was promising locally but failed causal confirmation.
B18A	grounded explanation	official MF1=.55507, Δ =.00976 vs B1	Main frozen P1 ensemble; validation-selected heterogeneous composition.
B18B	selective evidence/routing	frozen validation MF1=.711205; strict-test artifact pending	Main mechanism ablation: grounded uncertainty can selectively correct NEI.
B19	disagreement KD	OOF MF1=.667607 vs .660788 control	Secondary single-student compression; raw official seed-87 MF1=.54940.

Table 2: Condensed experimental trajectory. Values originate from their respective development, OOF, or official protocols and must not be compared as one common leaderboard.

Variant	Acc.	Macro-F1	F1-S	F1-R	F1-NEI
A0 Direct-only	.700549	.692045	.720348	.769904	.585882
A1 Grounded-only	.706731	.701191	.704519	.780834	.618221
A2 Equal ensemble	.699863	.692773	.707071	.772608	.598639
A3 Symmetric confidence	.698489	.691095	.706145	.771930	.595211
A4 No asymmetric NEI	.710165	.704436	.707904	.781915	.623491
A5 Disagreement-only	.706731	.701191	.704519	.780834	.618221
A6 Max-confidence-only	.698489	.691095	.706145	.771930	.595211
A8 Full B18B	.714973	.711205	.703016	.793008	.637591

Table 3: Frozen validation component ablations (official validation is used only for evaluation of frozen policies). A8 is the full asymmetric policy at $\tau = .49$. A5 equals A1 and A6 equals A3 on this prediction set; their inclusion documents the observed degeneracy. F1-S/R denote Supported/Refuted.

point-estimate comparison only: AMuFC predictions are unavailable, so no paired test can be performed. B19 disagreement distillation reaches Macro-F1 0.54940 with one student model. These numbers were produced after validation-only freezing.

The test-peak $\tau = 0.60$ result (Macro-F1 0.56166) is exploratory and is not used as the main claim because the threshold was identified after inspecting test outcomes. Similarly, any B18B row whose canonical prediction file is not yet reconciled remains TODO.

7 Additional Ablations and Negative Results

7.1 Sentence-selection branch

The repository contains a separately named B18B sentence-selector study; to avoid ambiguity, we call it *B18B-Sel* in this paper. B18B-Sel uses rationale-distilled cross-encoder scores to adaptively select one to three sentences. It is useful as an evidence-selection ablation, but it is *not* the dual-expert routing method in Figure 1. On Fold 0, rationale-distilled selection improves a generic cross-encoder by +.00865 MF1 but is still -.00214 behind the retrieval Top-3 control. Consequently it was closed after the preregistered screen rather than expanded to more seeds or test evaluation (Table 8). This negative result guards against attributing B18A gains to sentence compression alone.

7.2 Why earlier constraint approaches were not promoted

Earlier B6-B17 interventions supplied useful diagnostics but did not establish a stable main method. B6-A’s raw auxiliary ensemble reached .7080 MF1 on validation, yet its bootstrap probability against the compute-matched direct control was .9354, below the preregistered .95 gate. B9 and B13 exhibited strong Fold-0 gains but degraded under fresh-fold confirmation. B16’s counterfactual verdict curriculum improved Fold 0 by +.01654 MF1 versus its anchor, but aggregated confirmation was

System	Venue/year	Evidence and evaluation	Acc.	Reported F1	Comparability
MOCHEG (Yao et al., 2023)	SIGIR 2023	system text+image	–	F-score .4406	original benchmark; score label retained
LVL4M4FV (Tahmasebi et al., 2024)	arXiv 2024	retrieved text+image	.4510	micro-F1 .4510	macro not source-reported
HGTMFC (Pang et al., 2025)	AAAI 2025	retrieved text+image, raw test	.4861	Macro-F1 .4678	aligned split; different model/retrieval
MetaSumPerceiver (Chen et al., 2024b)	ACL 2024	system text+image	–	F-score .4860	source does not label it
MEVER (Zhang et al., 2026)	arXiv 2026	retrieved multimodal	.483±.021	MF1 .497±.012	Macro-F1 custom consistent preprocessing
M-RAV (Qwen2.5-32B) (Luu et al., 2027)	IPM 2027	system evidence, filtered $n = 2001$	.5002	MF1 .5014	filtered test; not aligned
Entailed Opinion (Kumar et al., 2025)	arXiv 2025	domain-generalization setting	–	MF1 .5700	not P1 system retrieval
AMuFC (Jung et al., 2026)	HCAI/CIKM 2025; arXiv 2026	retrieved text+image, raw test	.5460	MF1 .5400	aligned raw P1 challenger
GraphCURE-B1	this work	fixed corpus, raw P1 $n = 2442$	.5680	MF1 .5453	aligned raw P1
GraphCURE-B18A seed 100	this work	fixed corpus, raw P1 $n = 2442$	.5696	MF1 .5513	aligned raw P1, one seed
GraphCURE-B18A (8-model)	this work	validation-selected, raw P1 $n = 2442$	.5754	MF1 .5551	aligned raw P1, confirmatory
GraphCURE-B18B	this work	frozen routing, strict $n = 2434$	.5666	MF1 .5500	strict-only artifact; see TODO

Table 4: Protocol-aware comparison with prior work. “Aligned” means raw MOCHEG P1 test with system-retrieved evidence; it does not assert identical retrievers or model capacity. The table has more than ten prior/system rows, but only aligned rows may be used for a direct point-estimate comparison.

Evidence policy	Acc.	Grounded MF1	B18B MF1
$K = 1$	.707418	.660969	.704452
$K = 3$	.718407	.695423	.715270
$K = 5$	.714973	.701191	.711205
Direct-only	.700549	–	.692045

Table 5: Frozen evidence-count sensitivity on validation. $K = 3$ is selected by validation Macro-F1. The grounded-only optimum at $K = 5$ does not imply the routed-system optimum.

τ	Val. MF1	Raw-test MF1	Δ vs B1
.40	.70774	.55126	+0.00595
.45	.70932	.55428	+0.00897
.49 (validation-selected)	.71121	.55488	+0.00958
.50	.71109	.55577	+0.01046
.55	.71089	.55745	+0.01214
.60 (post-hoc peak)	.71012	.56166	+0.01635
.65	.70755	.55938	+0.01407
.70	.70420	.55627	+0.01096

Table 6: Threshold sensitivity. Only $\tau = .49$ is selected without test labels. The $\tau = .60$ raw-test maximum is exploratory and is not a confirmatory result.

GraphCURE system	Macro-F1	Accuracy	Raw-P1 status
B1 five-seed	.5453	.5680	frozen baseline
B18A seed 42	.5338	.5524	one seed
B18A seed 87	.5394	.5606	one seed
B18A seed 100	.5513	.5696	best single
B18A full 10-member	.5512	.5717	grounded seed
B18A deferral ($\tau = .49$)	.5549	.5713	unpruned control
B18A super-ensemble	.5551	.5754	frozen threshold
			main
			confirmatory
			result
B19 DKD seed 87	.5494	.5676	raw P1

Table 7: Raw official P1 test ($n = 2,442$) for frozen GraphCURE systems. The main result is the B18A heterogeneous ensemble selected on validation. B18B is not included because its available frozen routing output is a strict-track artifact.

Evidence policy	Acc.	MF1	F1-S	F1-NEI
Original Top-5 anchor	.6664	.6435	.6026	.5067
Retrieval Top-3	.6720	.6482	.5964	.5187
Generic CE selector	.6496	.6374	.5674	.5253
Rationale-distilled selector	.6651	.6460	.6098	.5103

Table 8: B18B-Sel train-only Fold-0 screen ($n = 2,326$). The rationale-distilled selector is better than generic CE but fails the promotion gate against retrieval Top-3.

-.00104 versus the matched control. These negative outcomes motivate the B18 design choice: preserve a competent direct decision model and use grounded supervision as a selective uncertainty signal rather than continuously modifying the anchor’s decision boundary.

8 Efficiency and Robustness Analysis

8.1 Inference efficiency

On an RTX 5090 with 256 common validation claims, batch size 4, and two warm-up batches, direct, grounded, B18B, and B19 systems require

320.72, 193.17, 513.88, and 64.47 ms per claim, respectively (Table 9). B19 is approximately 7.97 times faster than B18B. All ensemble timings are sequential; peak allocated memory is therefore a per-checkpoint peak, not a simultaneous eight-model memory requirement. Subset effectiveness scores produced by this benchmark are diagnostic only.

System	Passes	ms/claim	claims/s	Peak GiB
Direct ensemble	5	320.72	3.118	14.74
Grounded ensemble	3	193.17	5.177	14.74
B18B dual ensemble	8	513.88	1.946	14.74
B19 single student	1	64.47	15.511	14.74

Table 9: Efficiency benchmark: RTX 5090, 256 claims, batch size 4, two warm-up batches. Model-loading time is excluded.

Group	n	Δ MF1	Helpful/Harmful
Overall	1,456	+.01916	58 / 37
PolitiFact	431	-.00249	0 / 6
Snopes	1,025	+.01994	58 / 31
Gold NEI	478	+.03239	58 / 0
Gold Refuted	487	-.00343	0 / 9
Gold Supported	491	-.01404	0 / 28
Gold rank 1	1,233	+.02772	54 / 25
Gold rank 2–5	131	-.02607	3 / 7
Gold not retrieved	59	-.03260	1 / 4

Table 10: Selected frozen-validation subgroup audit: B18B minus direct. Gold rank is a diagnostic label unavailable at inference, not a routing input.

8.2 Subgroup behavior

On the frozen validation audit, B18B improves Macro-F1 by 1.916 points, with 58 helpful and 37 harmful changes. Table 10 separates where this gain occurs. It is largest for NEI (+3.239), rank-one evidence (+2.772), evidence hits (+2.280), and Snopes (+1.994). Retrieval misses decrease by 3.223 points and Supported claims decrease by 1.404 points. These results suggest that the method is an evidence-sufficiency correction, not a universal replacement for direct verification.

8.3 B19 distillation

Five-fold train-only OOF confirmation gives B19 control Macro-F1 0.660788 and disagreement-KD Macro-F1 0.667607, a +0.006819 difference. Four of five folds improve and paired bootstrap $P(\Delta > 0) = 0.9517$. Table 11 shows why the disagreement target matters: naive ensemble KD is worse than the matched control, while counterfactual loss alone is nearly neutral and the full objective reduces the screen gain. This supports B19 as a practical compression of complementary expertise, while B18B remains the main routing contribution.

9 Error Analysis

The dominant helpful transition is direct Refuted/Supported to grounded NEI on gold NEI claims. Harmful transitions are direct Supported to NEI (28 cases) and direct Refuted to NEI (9 cases). This asymmetry explains why B18B improves NEI

Fold-0 variant	Acc.	MF1	F1-S	F1-NEI
Matched control	.6690	.6570	.6013	.5368
Naive ensemble KD	.6707	.6513	.6106	.5152
Counterfactual only	.6758	.6577	.6063	.5332
Full KD + disagree + CF	.6776	.6584	.6022	.5359
Disagreement KD	.6836	.6687	.6276	.5472

Table 11: B19 train-only Fold-0 mechanism screen ($n = 2,326$). Full B19 confirmation uses only disagreement KD after the screen.

while slightly reducing decisive-class F1. The failure atlas also shows that routing under missing or rank-2–5 evidence is less reliable. A future system should estimate retrieval reliability before allowing deferral, but using gold rank or QREL status at inference would be invalid.

10 Discussion

The results support a narrow interpretation of the contribution. Evidence-grounded explanations are useful as a sufficiency sensor, but explanation supervision alone is not consistently beneficial: a matched explanation-loss ablation produced approximately zero aggregate gain. The useful object is the asymmetric interface between epistemic sensing and verdict prediction. This interface also exposes a compute–quality trade-off: B18B uses eight sequential expert passes, whereas B19 compresses the complementarity into one student.

11 Limitations and Ethical Considerations

MOCHEG is a single benchmark with two principal sources and potentially duplicated claims; we therefore report raw and strict tracks. The fixed-corpus protocol does not measure live web robustness. The grounded expert inherits teacher errors and pseudo-label noise. Subgroup analyses are descriptive and do not justify post-hoc policy tuning. The current repository contains inconsistent B18B test summaries; the final paper must replace the TODO row with a checksum-matched canonical evaluation. Automated fact checking can amplify source and retrieval bias; predictions should be treated as decision support, not authoritative adjudication.

12 Conclusion

We presented GraphCURE-B18B, a dual-expert asymmetric epistemic-deferral policy for evidence-grounded fact checking. On frozen validation it improves Macro-F1 by 1.92 points with strong

paired-bootstrap evidence, and subgroup analysis attributes the gain to correcting NEI decisions when high-quality evidence is available. B19 demonstrates that complementary experts can be compressed into a single efficient student. The remaining priority is canonicalizing the B18B official-test artifact and evaluating frozen policies on an external dataset before making a broad cross-domain claim.

Acknowledgments

[TODO: Add authors, affiliations, funding, and reproducibility URL after unblinding.]

References

- Isabelle Augenstein, Christina Lioma, Dongsheng Wang, Lucas Lima, Casper Hansen, Christian Hansen, and Jakob Simonsen. 2019. [Multifc: A real-world multi-domain dataset for evidence-based fact checking of claims](#). In *Proceedings of EMNLP-IJCNLP*, pages 4684–4696.
- Jifan Chen, Grace Kim, Aniruddh Sriram, Greg Durrett, and Eunsol Choi. 2024a. [Complex claim verification with evidence retrieved in the wild](#). In *Proceedings of NAACL-HLT*, pages 3569–3587.
- Ting-Chih Chen, Chia-Wei Tang, and Chris Thomas. 2024b. [MetaSumPerceiver: Multimodal multi-document evidence summarization for fact-checking](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8742–8757.
- Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. [A survey on automated fact-checking](#). *Transactions of the ACL*, 10:178–206.
- Andreas Hanselowski, Hao Zhang, Zhijiang Li, Daniil Sorokin, Benjamin Schiller, Claudia Schulz, and Iryna Gurevych. 2019. Ukp-athene: Multi-sentence textual entailment for fact checking. In *Proceedings of the FEVER Workshop*, pages 103–108.
- Jaeyoon Jung, Yejun Yoon, and Kunwoo Park. 2026. Is a picture worth a thousand words? adaptive multi-modal fact-checking with visual evidence necessity. *arXiv preprint arXiv:2604.04692*.
- Gaurav Kumar, Ayush Garg, Debajyoti Mazumder, Aditya Kishore, Babu Kumar, and Jasabanta Patro. 2025. Entailed opinion matters: Improving the fact-checking performance of language models by relying on their entailment ability. *arXiv preprint arXiv:2505.15050*.
- Son T. Luu, Trung Vo, and Le-Minh Nguyen. 2027. [M-RAV: Multimodal retrieve-augment-verify framework for boosting zero-shot fact verification system with large language models](#). *Information Processing & Management*, 64(1):104988.
- Hui Pang, Chaozhuo Li, Litian Zhang, Senzhang Wang, and Xi Zhang. 2025. [Beyond text: Fine-grained multi-modal fact verification with hypergraph transformers](#). In *Proceedings of AAAI*, volume 39, pages 6389–6397.
- Sahar Tahmasebi, Eric Müller-Budack, and Ralph Ewerth. 2024. Multimodal misinformation detection using large vision-language models. *arXiv preprint arXiv:2407.14321*.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. [Fever: A large-scale dataset for fact extraction and verification](#). In *Proceedings of NAACL-HLT*, pages 809–819.
- William Yang Wang. 2017. [Liar, liar pants on fire: A new benchmark dataset for fake news detection](#). In *Proceedings of ACL*, pages 422–426.
- Barry Menglong Yao, Aditya Shah, Lichao Sun, Jin-Hee Cho, and Lifu Huang. 2023. [End-to-end multimodal fact-checking and explanation generation: A challenging dataset and models](#). In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2733–2743.
- Zhenrui Yue, Huimin Zeng, Lanyu Shang, Yifan Liu, Yang Zhang, and Dong Wang. 2024. Retrieval augmented fact verification by synthesizing contrastive arguments. In *Proceedings of ACL*.
- Delvin Ce Zhang, Suhan Cui, Zhelin Chu, Xianren Zhang, and Dongwon Lee. 2026. MEVER: Multi-modal and explainable claim verification with graph-based evidence retrieval. *arXiv preprint arXiv:2602.10023*.
- Liwen Zheng, Chaozhuo Li, Xi Zhang, Yuming Shang, Feiran Huang, and Haoran Jia. 2024. Evidence retrieval is almost all you need for fact verification. In *Findings of ACL*, pages 9274–9281.

A Detailed Development Ledger

Table 12 expands the main trajectory with the numerical evidence used to close or advance each hypothesis. These are reported for reproducibility and to prevent failed development screens from being misrepresented as official-test improvements.

B Five-Fold B19 Confirmation

C Raw versus Strict Test Reporting

The raw official test ($n = 2,442$) is the primary literature-comparison track. The strict test ($n = 2,434$) removes eight verbatim cross-split duplicates and is a robustness track only. Table 14 does not merge the two populations. The B18B strict routing output has an alignment/intersection artifact documented in project logs; until a checksum-matched re-evaluation produces a raw and strict pair, its result remains a robustness diagnostic rather than a headline test result.

D Reproducibility Checklist

The repository records manifests, retrieval signatures, prediction files, training summaries, seeds, and protocol flags. The main scripts are ‘train_mocheg_qwen3_lora_verifier.py’, ‘train_mocheg_b18_explanation_verifier.py’, ‘analyze_mocheg_b18b_component_ablations.py’, ‘analyze_mocheg_b18b_subgroups.py’, ‘train_mocheg_b19_distillation.py’, and ‘benchmark_mocheg_b18b_efficiency.py’. Before submission, add exact GPU driver versions, package-lock information, canonical B18B raw/strict prediction hashes, author metadata, and a public archival repository link.

ID	Intervention	Quantitative observation	Interpretation
B2-SV	flat versus hierarchical sufficiency verification	Flat MF1=.6371; SV=.6398 (+.0027) on Fold 0; five-fold confirmation $+.00060 \pm .00745$	Hierarchy had small local benefit but no stable promotion.
B3	source/evidence DRO	qrel-absent MF1 +.1004; overall MF1 -.0021	Robustness improvement cannot be bought with aggregate regression.
B4	atomic sentence verifier	sentence retrieval Recall@8=.8661; verdict MF1 -.0270 vs article evidence	Retrieval coverage alone is insufficient; discourse matters.
B5	context packets	best interpolation $+.01014$ on one validation split; five-seed packet gain $+.00387 \pm .00862$	Complementarity exists but is not a reliable replacement.
B6-A	sufficiency/polarity multi-task	auxiliary vs matched control $+.00886 \pm .01081$; bootstrap .9354	Positive transfer signal, below preregistered stability gate.
B6-B	PCGrad	conflict rate about 45–46% in epoch 1; ensemble -.0105 vs control	Gradient conflict was real but projection did not improve generalization.
B7	selective expert router	MF1 $+.00460$; bootstrap .9294	Helpful precision, insufficient gain and confidence.
B8	class-logit bias	MF1 $+.00216$; bootstrap .7604	Prior shift does not explain the main error pattern.
B9	fixed seed ensemble	Fold-0 $+.01348$; fresh-fold mean $+.00398$	A screen peak is not confirmation.
B10	cross-fitted calibrator	aggregate MF1=.67054; $+.00345$ vs unweighted ensemble	Bootstrap/source gates fail.
B11	provenance calibrator	-.00176 vs unweighted ensemble	Provenance-conditioned calibration harms aggregate accuracy.
B12	joint constraints from base	$+.00538$ vs compute control but $-.01376$ vs anchor	Constraint tasks help relative to compute but induce anchor drift.
B13	curriculum/blend	Fold-0 blend $+.00510$; five-fold mean $+.00090$	Direct curriculum is unstable.
B14	NEI escape analysis	MF1 $+.00209$; accuracy $+.01126$	It helps supported/refuted escape but damages NEI under qrel absence.
B15	value-of-information gate	MF1 $+.00380$; AUROC=.5660	Observable utility is too weak for reliable routing.
B16	counterfactual omission	Fold-0 $+.01654$ vs anchor; confirmation $-.00104$ vs control	The strongest screen still fails matched-control confirmation.
B17	failure atlas	candidate/control .66517/.66621; help/harm 550/561	No stable subgroup justified further tuning on inspected folds.

Table 12: B2–B17 negative and diagnostic experiments. Each value remains in its native protocol; no row is a P1 official-test claim.

Fold	Control MF1	DKD MF1	Δ
0	.657035	.668656	$+.011621$
1	.659998	.656767	$-.003231$
2	.662265	.667143	$+.004878$
3	.647130	.660188	$+.013058$
4	.674931	.681645	$+.006714$
Aggregate	.660788	.667607	$+.006819$

Table 13: B19 train-only out-of-fold confirmation. Four of five folds improve; the aggregate paired bootstrap probability is .9517.

System	Track	Acc.	MF1
B1 five-seed	raw official	.56798	.54531
B1 five-seed	strict	.56902	.54581
B18A 8-model	raw official	.57535	.55507
B18A 8-model	strict	.57477	.55383
B19 seed 87	raw official	.56757	.54940
B19 seed 87	strict	.56615	.54733
B18B frozen router	strict intersection	.56656	.54998

Table 14: Track-separated test results. The B18B result is not used for the main official-test claim.